



# Girraween High School

## Year 12 Mathematics Extension 2

### Task 1, December 2024

#### General Instructions

- Reading Time - 5 minutes
- Working Time - 90 minutes
- Write using black pen only
- Calculators approved by NESA may be used
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total Marks: 66**

#### Section I - 5 marks

- Attempt questions 1-5
- Allow about 10 minutes for this section

#### Section II - 61 marks

- Attempt questions 6-11
- Allow about 80 minutes for this section

### Section I

5 marks

Attempt questions 1 - 5

Allow about 10 minutes for this section

Use the multiple choice answer sheet for Questions 1 to 5.

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#### Question 1

A quadratic equation with zeros of  $4 + i$  and  $4 - i$  is

(A)  $x^2 - 8x + 17 = 0$

(B)  $x^2 + 8x + 17 = 0$

(C)  $x^2 - 8x - 17 = 0$

(D)  $x^2 + 8x - 17 = 0$

#### Question 2

The complex number  $w$  is a root of the equation  $z^3 + 1 = 0$ .

Which of the following is FALSE?

(A)  $\bar{w}$  is also a root

(B)  $w^2 + 1 - w = 0$

(C)  $\frac{1}{w}$  is also a root

(D)  $(w - 1)^3 = -1$

#### Question 3

$P(z)$  is a polynomial in  $z$  of degree 4 with real coefficients.

Which one of the following statements must be false?

(A)  $P(z)$  has four real roots.

(B)  $P(z)$  has two real roots and two non-real roots.

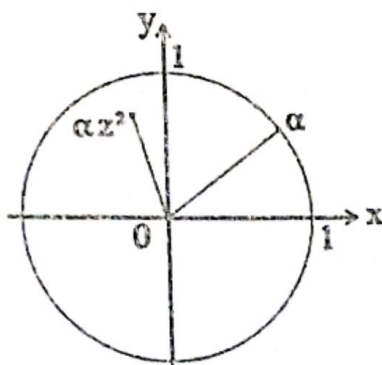
(C)  $P(z)$  has one real root and three non-real roots.

(D)  $P(z)$  has no real roots.

*Examination continues next page...*

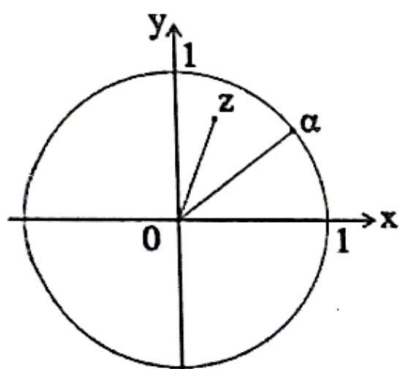
#### Question 4

The Argand diagram below shows the complex numbers  $\alpha$  and  $\alpha z^2$ .

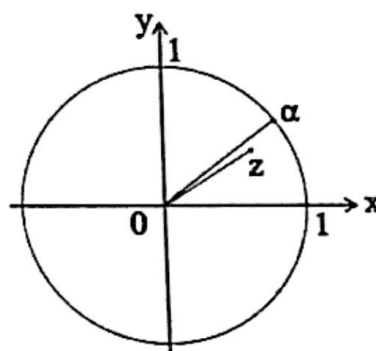


Which of the following best represents the positions of  $\alpha$  and  $z$ ?

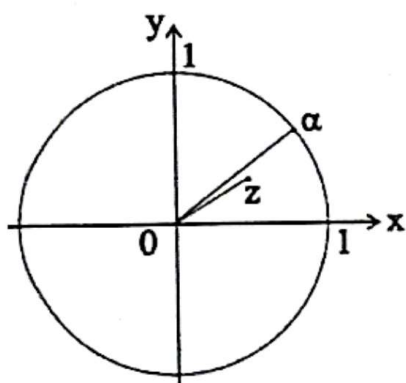
A)



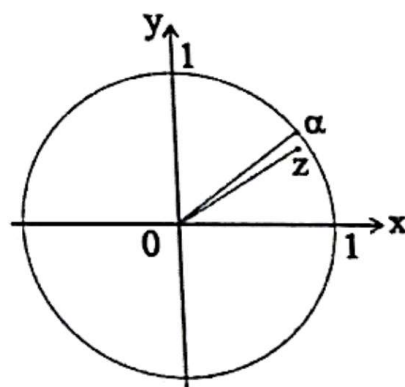
B)



C)



D)



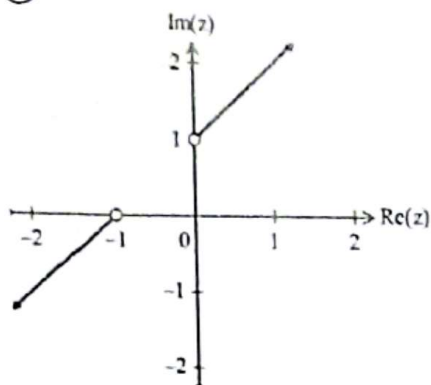
*Examination continues next page...*

Question 5

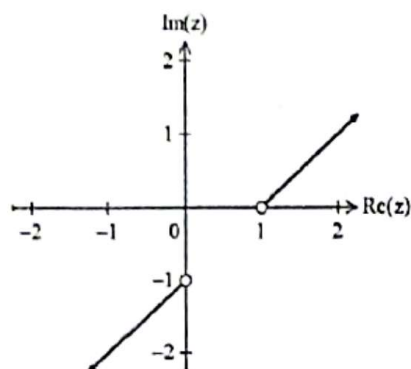
Which of the following Argand diagrams describes the locus defined by

$$\arg(z - i) = \arg(z + 1)?$$

(A)

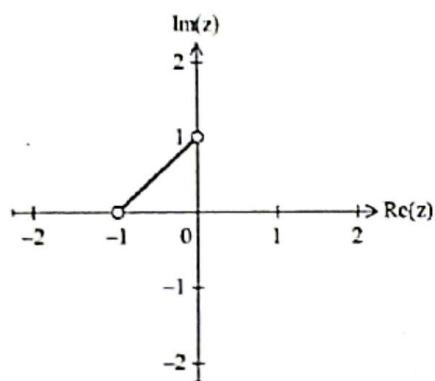


B)

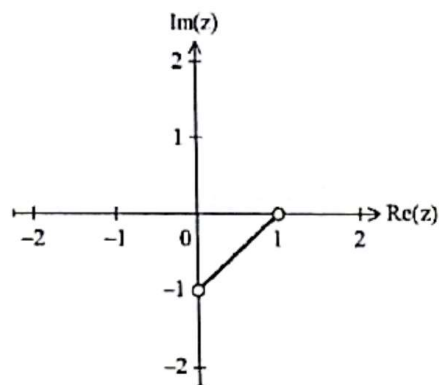


$$(z - i) = (z + 1)$$

C)



D)



*End of Section 1*

*Examination continues next page...*

## Section II

61 marks

Attempt questions 6 - 11

Allow about 80 minutes for this section

Start each question on a new page.

In Questions 6 - 11, your responses should include relevant mathematical reasoning and/or calculations.

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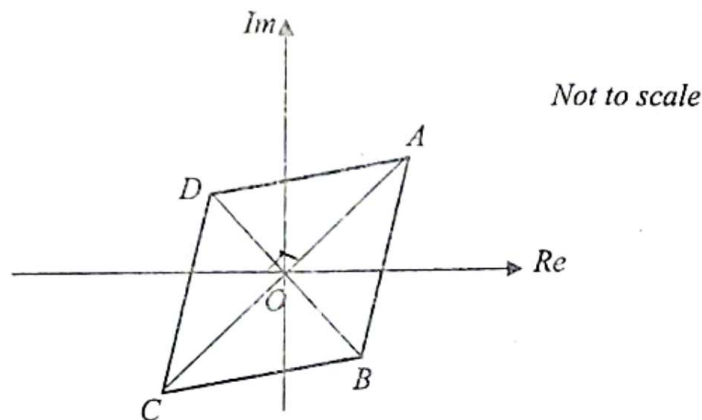
| Question 6 (11 marks) Start a new answer page                                                                                     | Mark |
|-----------------------------------------------------------------------------------------------------------------------------------|------|
| a) Given that $z_1 = 3 - i$ , $z_2 = 2 + 5i$ , express in the form, $a + ib$ , where $a$ and $b$ are real                         |      |
| i) $(\overline{z_1})^2$                                                                                                           | 1    |
| ii) $\frac{z_1}{z_2}$                                                                                                             | 1    |
| iii) $\left  \frac{z_1}{z_2} \right $                                                                                             | 1    |
| b) i) Express $1 - i\sqrt{3}$ in modulus-argument form.                                                                           | 1    |
| ii) Hence evaluate $(1 - i\sqrt{3})^6$ .                                                                                          | 1    |
| c) If $w$ is one of the complex roots of $z^3 - 1 = 0$ , find the value of $\frac{1}{1+w} + \frac{1}{1+w^2}$ .                    | 2    |
| d) i) Find the complex square roots of $-18i$ , expressing your answer in the form $a + bi$ , where $a$ and $b$ are real numbers. | 2    |
| ii) Hence, solve the quadratic equation $z^2 - 3(1 + i)z + 9i = 0$                                                                | 2    |

*Examination continues next page...*

Question 7 (9 marks) Start a new answer page

Mark

- a) In the diagram below,  $ABCD$  is a rhombus whose diagonals meet at the origin,  $O$ .



If  $A$  represents the complex number  $z_1 = 3 + 4i$  and  $OD = 2$  units, find the complex number (in Cartesian form) represented by

- |                                    |   |
|------------------------------------|---|
| i) $D$ (let this point be $z_2$ )  | 2 |
| ii) $C$ (let this point be $z_3$ ) | 1 |
- b) The complex number  $z$  and its conjugate  $\bar{z}$  satisfies  $(z - \bar{z})^2 + 8(z + \bar{z}) = 16$ . 2
- Find the Cartesian equation of the locus.
- c) i) Find the complex sixth roots of unity in modulus-argument form. 2
- ii) Sketch the roots on an Argand diagram. 2

*Examination continues next page...*

**Question 8 (12 marks) Start a new answer page**

**Mark**

a) i) Given  $\sqrt{3} + i = 2\text{cis}\frac{\pi}{6}$ , state  $\sqrt{3} - i$  in modulus argument form.

1

ii) Hence, find the smallest positive integer  $m$ , such that  $(\sqrt{3} + i)^m = (\sqrt{3} - i)^m$ .

3

b) Consider the polynomial  $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 12$ .

4

Given  $x = 1 - i\sqrt{2}$  is a root of  $P(x) = 0$ , solve the equation  $P(x) = 0$ .

c) Let  $z^n = \cos n\theta + i \sin n\theta$

i) Show that  $\sin n\theta = \frac{z^n - z^{-n}}{2i}$  and  $\cos n\theta = \frac{z^n + z^{-n}}{2}$ .

2

ii) Hence, or otherwise prove the identity

2

$$32 \sin^4 \theta \cos^2 \theta = \cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2.$$

$$\sin^4 \theta (1 - \cos^2 \theta)^2$$

**Question 9 (10 marks) Start a new answer page**

**Mark**

a) (i) Draw a sketch of the locus of  $z$  given by the equation

2

$$\arg(z - 1) - \arg(z + 1) = \frac{\pi}{4}.$$

(ii) Find the centre of the circle on which this locus lies.

2

b) i) By considering  $(\cos\theta + i\sin\theta)^5$  using De Moivre's Theorem prove that

3

$$\sin 5\theta = \sin\theta(5 - 20\sin^2\theta + 16\sin^4\theta)$$

ii) By letting  $\theta = \frac{\pi}{5}$ , show that the exact value of  $\sin^2(\frac{\pi}{5})$  is  $\frac{1}{8}(5 - \sqrt{5})$ .

3

*Examination continues next page...*

**Question 10 (11 marks) Start a new answer page**

**Mark**

- a) The complex number  $z$  satisfies both  $|z - 1| \leq |z - i|$  and  $|z - 2 - 2i| \leq 1$ .
- In an Argand diagram, indicate the region which contains the point  $P$  representing  $z$ . 3
  - If  $P$  moves on the boundary of this region and  $\arg(z - 1) = \frac{\pi}{4}$ , find the value of  $z$  in the form  $x + iy$  where  $x$  and  $y$  are real. 3
- b) Let  $z = e^{i\theta}$ . Consider the expression  $S_n = 1 + z + z^2 + \dots + z^n$ .
- Show that  $S_n = \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1}$  1
  - Show that  $S_n = e^{\frac{in\theta}{2}} \times \frac{(\sin \frac{n+1}{2} \theta)}{\sin(\frac{\theta}{2})}$  2
  - Deduce that  $\sin \theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta = \frac{\sin \frac{n\theta}{2} \sin \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}$  2

**Question 11 (8 marks) Start a new answer page**

**Mark**

Consider the polynomial  $z^5 - i = 0$ .

- Find the roots of  $z^5 = i$  2
- Show that  $(z - i) \left( z^2 - 2i \sin \frac{\pi}{10} z - 1 \right) \left( z^2 + 2i \sin \frac{3\pi}{10} z - 1 \right) = 0$  4
- Given  $z^5 - i = (z - i)(z^4 + iz^3 - z^2 - iz + 1)$ , show that  $\sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$ . 2

*End of examination.*

# Maths Ext 2 Solutions

## 2024-25 Task 1

### Multichoice

Q1  $(x-4-i)(x-4+i)=0$   
 $[(x-4)-i][(x-4)+i]=0$   
 $(x-4)^2 - i^2 = 0$   
 $(x-4)^2 + 1 = 0$   
 $x^2 - 8x + 16 + 1 = 0$   
 $x^2 - 8x + 17 = 0$  (A)

Q2 The roots of  $z^3 + 1 = 0$

are  $\omega$ ,  $\omega^2$  and  $-1$ .

sum of roots  $\omega + \omega^2 + (-1) = 0$

which is equivalent to

$$\omega^2 + 1 - \omega = 0$$

(D) is false

Q3 A and D are possible.

however no real roots occur  
in conjugate pairs since  
coefficients are real.

$\therefore$  (C) is false.

Q4 Since  $|z| < 1$

then  $|z|^2 < |z| < 1$

and  $\arg(\alpha z^2) = 2\arg(z) + \arg(\alpha)$

(D)

Q5

(A)

Q6

$$\begin{aligned} \text{a) i) } (\bar{z})^2 &= (3+i)^2 \\ &= 9 + 6i + i^2 \\ &= 8 + 6i \end{aligned}$$

*very well done*

(1)

$$\text{ii) } \frac{z_1}{z_2} = \frac{3-i}{2+5i} \times \frac{2-5i}{2-5i}$$

$$= \frac{6 - 15i - 2i - 5}{4 - 25i^2}$$

*very well done*

$$= \frac{1}{29} - \frac{17i}{29}$$

(1)

$$\text{iii) } \left| \frac{z_1}{z_2} \right| = \left| \frac{1}{29} - \frac{17i}{29} \right|$$

$$= \sqrt{\left(\frac{1}{29}\right)^2 + \left(\frac{17}{29}\right)^2}$$

$$= \sqrt{\frac{10}{29}}$$

*most students have well done it some students forgot to find modulus.*

(1)

$$\text{OR } \left| \frac{z_1}{z_2} \right| = \left| \frac{3-i}{2+5i} \right| = \frac{\sqrt{3^2 + 1^2}}{\sqrt{2^2 + 5^2}}$$

$$= \sqrt{\frac{10}{29}}$$

Q6 b) i)  $1 - \sqrt{3}i$

Now  $|1 - \sqrt{3}i| = \sqrt{1^2 + \sqrt{3}^2} = 2$

$\arg(1 - \sqrt{3}i) = \tan^{-1}(-\sqrt{3})$   
 $= -\frac{\pi}{3}$  *very well done*

$\therefore 1 - \sqrt{3}i = 2 \operatorname{cis}\left(-\frac{\pi}{3}\right)$  (1)

ii)  $(1 - \sqrt{3}i)^6 = \left[2 \operatorname{cis}\left(-\frac{\pi}{3}\right)\right]^6$

$= 2^6 \operatorname{cis}\left(-\frac{6\pi}{3}\right)$  *very well done*

$= 64$  (1)

*Most students have shown excellent understanding of Q6 except for some calculation mistakes.*

c)

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\therefore \omega^3 = 1$$

$$\omega^2 + \omega + 1 = 0$$

①

(since  $\omega$  is complex)

$$\frac{1}{1+\omega} + \frac{1}{1+\omega^2} = \frac{1+\omega^2+1+\omega}{(1+\omega)(1+\omega^2)}$$

$$= \frac{1+\omega^2+1+\omega}{1+\omega^2+\omega+\omega^3}$$

$$= \frac{1+0}{0+1}$$

$$\text{since } \omega^2 + \omega + 1 = 0$$

$$\omega^3 = 1$$

$$= 1$$

①

Students have shown good understanding of this concept. Most students used  $\omega^3 = 1$  and  $\omega^2 + \omega + 1 = 0$  but a few number of students found the roots and substituted them.

(Qd)  $z^2 = -18i$

let  $z = a + ib$

$$(a + ib)^2 = -18i$$

$$a^2 + 2abi + b^2 = -18i$$

equating real & imaginary

$$a^2 - b^2 = 0 \quad (1)$$

$$2ab = -18$$

$$a = -\frac{9}{b} \quad (2)$$

Subs (2) into (1)

$$\left(-\frac{9}{b}\right)^2 - b^2 = 0$$

$$\frac{81}{b^2} - b^2 = 0$$

$$b^4 = 81$$

$$b = \pm 3$$

$$\therefore a = -\frac{9}{\pm 3} = \mp 3$$

$\therefore$  The square roots are  $3 - 3i$  and  $-3 + 3i$

or  $\pm 3 \mp 3i$

Very well done  
Almost all students  
have shown under-  
standing or finding  
square roots

$$\text{Q6 d(ii)} \quad z^2 - 3(1+i)z + 9i = 0$$

$$z = \frac{3(1+i) \pm \sqrt{9(1+i)^2 - 4(1)(9i)}}{2 \cdot 1}$$

$$= \frac{3 + 3i \pm \sqrt{18i - 36i}}{2}$$

$$= \frac{3 + 3i \pm \sqrt{-18i}}{2}$$

(1)

$$= \frac{3 + 3i \pm (3 - 3i)}{2} \quad \text{from (1)}$$

$$= \frac{6}{2} \quad \text{or} \quad \frac{6i}{2}$$

$$= 3 \quad \text{or} \quad 3i$$

(1)

very well done apart from a few calculation mistakes.

Q9 a) (i)  $|OA| = \sqrt{3^2 + 4^2} = 5$

$z_2$  is a  $\frac{\pi}{2}$  rotation

from  $z_1$ , and  $\frac{2}{5}$  the length

of  $z_1$ .

*Almost everyone could see the direction was  $-3+4i$ . However, many students didn't take into account the modulus of  $OA$  was 5 & didn't make  $OD = \frac{2}{5} OA$ .*

$$\begin{aligned} \therefore z_2 &= z_1 \times i \times \frac{2}{5} \\ &= (3 + 4i) \times i \times \frac{2}{5} \end{aligned}$$

$$z_2 = -\frac{8}{5} + \frac{6}{5}i$$

① multiplying  $z_1$  by  $i$

① multiplying  $z_1$  by  $\frac{2}{5}$

(ii)  $z_3$  is a rotation of  $\pi$  from  $z_1$ ,

$$\begin{aligned} \therefore z_3 &= -1 \times (3 + 4i) \\ &= -3 - 4i \end{aligned}$$

①

*Very simple.*

*Almost everyone was correct.*

Q75)

$$(z - \bar{z})^2 + 8(z + \bar{z}) = 16$$

let  $z = x + iy$

$$[x + iy - (x - iy)]^2 + 8[x + iy + (x - iy)] = 16$$

Now  $z + \bar{z} = 2\operatorname{Re}(z) = 2x$

$$z - \bar{z} = 2i\operatorname{Im}(z) = 2iy$$

$$(2iy)^2 + 8(2x) = 16$$

$$-4y^2 = 16 - 16x$$

$$y^2 = 4(x - 1)$$

Most students realised to let  $z = x + iy$  and also realised  $z - \bar{z} = 2iy$  &  $z + \bar{z} = 2x$ .

Some students forgot that  $(2iy)^2 = -4y^2$ ; NOT  $4y^2$ .

Q7c)

i)  $z^6 = 1$

let  $z = \text{cis } \theta$   
 $z^6 = (\text{cis } \theta)^6 = \text{cis } 6\theta$

now  $1 = \text{cis } 0$   
 equating arguments

① progress

Almost everyone got (i) right.  $6\theta = 0 + 2k\pi$   
 $\theta = \frac{k\pi}{3}$  for  $k = 0, \pm 1, \pm 2, 3$

$z_1 = \text{cis } 0 = 1$

$z_4 = \text{cis } \frac{2\pi}{3}$

$z_2 = \text{cis } \frac{\pi}{3}$

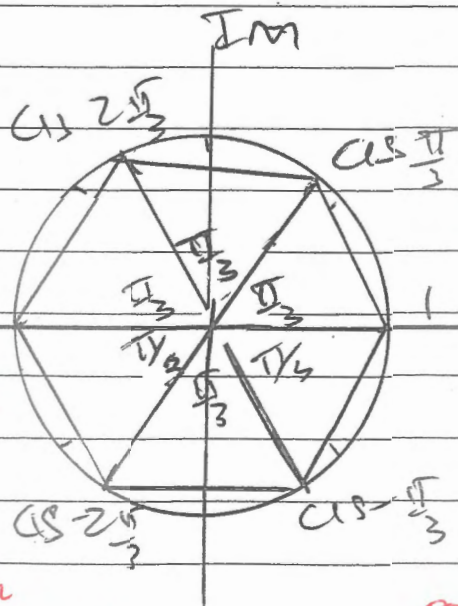
$z_5 = \text{cis } -\frac{2\pi}{3}$

$z_3 = \text{cis } -\frac{\pi}{3}$

$z_6 = \text{cis } \pi = -1$

①

(ii)



A simple question done with a LOT of careless omissions.

Students of ten didn't either (1) show the angles were  $\frac{\pi}{3}$

① shape or

① (labels) (2) show the moduli were all 1

Which SHOULD HAVE BEEN DONE BY PUTTING THEM ALL ON A UNIT CIRCLE!

Marking note:

(1) If they labelled  $\text{cis } \frac{\pi}{3}$  etc. I counted that as angle labels.

(2) if they labelled all of the moduli as 1

I didn't penalise them for not having the unit circle. [This was very rare].

Q 891 i)  $z = 2 \operatorname{cis} \frac{\pi}{6}$

$\therefore \bar{z} = 2 \operatorname{cis} \left(-\frac{\pi}{6}\right)$  (1)

well done

(ii)  $(\sqrt{3} + i)^m = (\sqrt{3} - i)^m$

$\left(2 \operatorname{cis} \frac{\pi}{6}\right)^m = \left(2 \operatorname{cis} -\frac{\pi}{6}\right)^m$

$2^m \operatorname{cis} \frac{m\pi}{6} = 2^m \operatorname{cis} -\frac{m\pi}{6}$  (1)

by De Moivre's

$\cos \frac{m\pi}{6} + i \sin \frac{m\pi}{6} = \cos \left(-\frac{m\pi}{6}\right) + i \sin \left(-\frac{m\pi}{6}\right)$

Now,  $\cos \frac{m\pi}{6} = \cos \left(-\frac{m\pi}{6}\right)$

and  $\sin \left(-\frac{m\pi}{6}\right) = -\sin \frac{m\pi}{6}$

$\therefore 2 i \sin \frac{m\pi}{6} = 0$  (1)

$\frac{m\pi}{6} = \pi$

$\therefore$  Smallest positive integer  $m$  is 6

(1) well done

Q 85)

Since the coefficients of the polynomial are real <sup>Must mention</sup>

$z = 1 + \sqrt{2}i$  is also a root. ①

$$[z - (1 + \sqrt{2}i)][z + (1 + \sqrt{2}i)]$$

$$\text{Now } z^2 - 2\operatorname{Re}(z) + |z|^2$$

$$= z^2 - 2z + (1^2 + \sqrt{2}^2)$$

$$= z^2 - 2z + 3 \text{ is a factor} \quad \text{①}$$

$$\begin{array}{r} z^2 - 2z + 4 \quad \text{①} \\ z^2 - 2z + 3 \overline{) z^4 - 4z^3 + 11z^2 - 14z + 12} \\ \underline{z^2 - 2z^3 + 3z^2} \phantom{+ 12} \\ -2z^3 + 8z^2 - 14z \phantom{+ 12} \\ \underline{-2z^3 + 4z^2 - 6z} \phantom{+ 12} \\ 4z^2 - 8z + 12 \phantom{+ 12} \\ \underline{4z^2 - 8z + 12} \\ 0 \end{array}$$

$$z^2 - 2z + 4 = 0$$

$$z = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(4)}}{2(1)}$$

well done

$$= \frac{2 \pm 2\sqrt{3}i}{2} = 1 \pm \sqrt{3}i \quad \text{①}$$

$$\therefore \text{ solutions are } z = 1 \pm \sqrt{2}i$$

$$z = 1 \pm \sqrt{3}i$$

Q 8c) i)

$$\text{Given } z^n = \cos n\theta + i \sin n\theta \quad (1)$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

① { since  $\cos(-n\theta) = \cos n\theta$   
and  $\sin(-n\theta) = -\sin n\theta$

$$\therefore z^{-n} = \cos n\theta - i \sin n\theta \quad (2)$$

$$z^n + z^{-n} = 2 \cos n\theta$$

$$\therefore \cos n\theta = \frac{1}{2} (z^n + z^{-n})$$

$$z^n - z^{-n} = 2i \sin n\theta$$

$$\therefore \sin n\theta = \frac{1}{2i} (z^n - z^{-n})$$

well done.

students realised if

$$\sin(-n\theta) = -\sin(n\theta)$$

not shown.

Q 8c) (ii)

$$RHS = 32 \sin^4 \theta \cos^2 \theta$$

$$\begin{aligned} &= 32 \sin^4 \theta (1 - \sin^2 \theta) \\ &= 32 \sin^4 \theta - 32 \sin^6 \theta \end{aligned}$$

$$\textcircled{1} \left\{ = 32 \left( \frac{1}{2i} \left( z - \frac{1}{z} \right) \right)^4 - 32 \left( \frac{1}{2i} \left( z - \frac{1}{z} \right) \right)^6 \right.$$

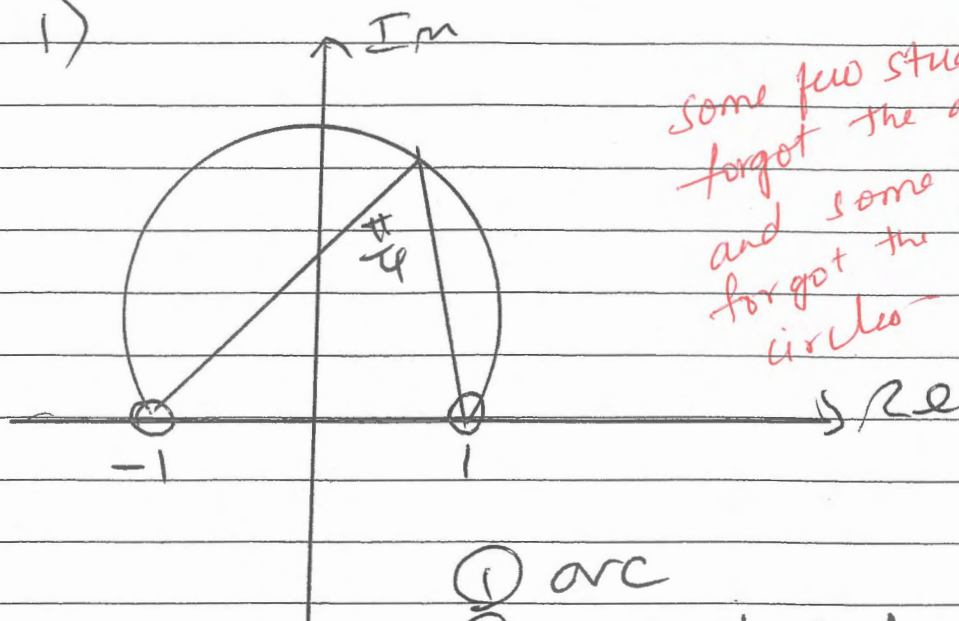
$$\begin{aligned} &= 2 \left( z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4} \right) \\ &\quad + \frac{1}{2} \left( z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} \right. \\ &\quad \left. - 6z^{-4} + z^{-6} \right) \\ &\quad = \frac{1}{2} \left( z^6 + \frac{1}{z^6} \right) - \left( z^4 + \frac{1}{z^4} \right) \\ &\quad \quad - 2 \left( z^2 + \frac{1}{z^2} \right) + 2 \end{aligned}$$

$$= \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2$$

$$= LHS$$

poorly done.

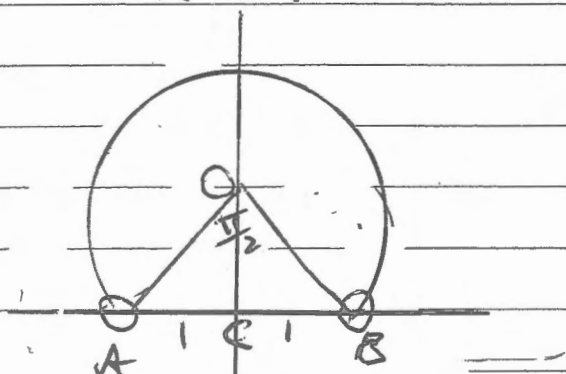
Q9 a) i)



some few students forgot the arc and some students forgot the open circles

- ① arc
- ① correct end pts

ii) centre lies on the vertical axis.

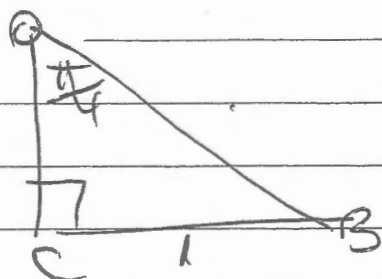


- ① reasoning
- ① centre

$$\angle AOB = \frac{\pi}{2}$$

Angle at the centre is twice the angle at the circumference.

$\triangle AOB$  is also isosceles



$$\tan \frac{\pi}{4} = \frac{1}{OC}$$

$$\therefore OC = 1$$

$\therefore$  centre  $(0, 1)$

very well done

Students to write  $C(0, 1)$

however not penalised

9.57 i)

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

by De Moivre's Theorem

1 5 10 10 5 1

$$(\cos \theta + i \sin \theta)^5$$

$$= \cos^5 \theta + 5 \cos^4 \theta i \sin \theta + 10 \cos^3 \theta i^2 \sin^2 \theta$$

$$\textcircled{1} \quad \begin{aligned} &+ 10 \cos^2 \theta i^3 \sin^3 \theta + 5 \cos \theta i^4 \sin^4 \theta \\ &+ i^5 \sin^5 \theta \end{aligned}$$

Expanding

$$= \cos^5 \theta + 5 \cos^4 \theta i \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 5 \cos \theta i \sin^3 \theta + i \sin^5 \theta$$

Equating imaginary parts

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta$$

$$\textcircled{2} \quad -10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$= \sin \theta (5 \cos^4 \theta - 10 \cos^2 \theta \sin^2 \theta + \sin^4 \theta)$$

apply  $\cos^2 \theta = 1 - \sin^2 \theta$ 

$$= \sin \theta [5 (1 - \sin^2 \theta)^2 - 10 (1 - \sin^2 \theta) \sin^2 \theta + \sin^4 \theta]$$

$$= \sin \theta [5 (1 - 2 \sin^2 \theta + \sin^4 \theta) - 10 \sin^2 \theta + 10 \sin^4 \theta + \sin^4 \theta]$$

So,  $\sin^2 \frac{\pi}{5} = \frac{5 + \sqrt{5}}{8}$  or  $\frac{5 - \sqrt{5}}{8}$

Now  $\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\sin^2 \frac{\pi}{4} = \frac{1}{2}$

$\sin^2 \frac{\pi}{5} < \sin^2 \frac{\pi}{4} < \frac{1}{2}$

$\therefore \sin^2 \frac{\pi}{5} = \frac{5 - \sqrt{5}}{8}$  (D)

Very well done

apart from justification

Most students who lost a mark  
as they did not give a convincing  
justification.

Q10

a) let  $z = x + iy$

$$|z-1| \leq |z-i|$$

$$|x+iy-1| \leq |x+iy-i|$$

$$|x-1+iy| \leq |x+1(y-1)|$$

$$\sqrt{(x-1)^2 + y^2} \leq \sqrt{x^2 + (y-1)^2}$$

$$x^2 - 2x + 1 \leq x^2 + y^2 - 2y + 1$$

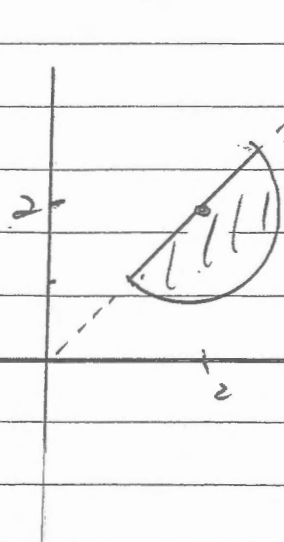
The vast majority of students did (10)(a)(i) well.  $y \leq x$

① or correct graph

$$|z - 2 - 2i| \leq 1$$

$$|z - (2 + 2i)| \leq 1$$

This is a circle with centre  $(2, 2)$  and radius 1



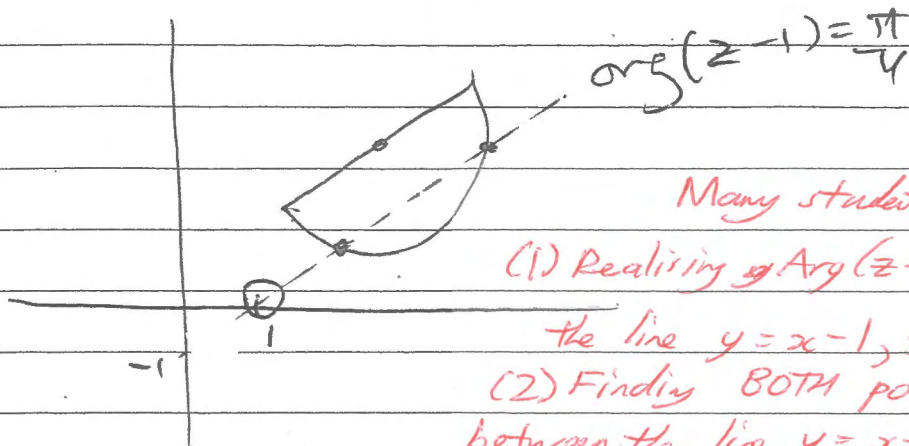
① correct region

(no penalty if  $y=x$  is all solid)

① or correct graph

ii) If  $\arg(z-1) = \frac{\pi}{4}$

then  $z$  lies on the line  $y = x - 1$  (1)



Many students had trouble

(1) Realising  $\arg(z-1) = \frac{\pi}{4}$  is

the line  $y = x - 1, x > 1$

(2) Finding BOTH points of intersection between the line  $y = x - 1$  & the circle.

Solve  $y = x - 1$  (1)

2  $(x-2)^2 + (y-2)^2 = 1$  (2)

simultaneously by

substituting (1) into (2)

$$(x-2)^2 + (x-1-2)^2 = 1$$

$$x^2 - 4x + 4 + x^2 - 6x + 9 = 1$$

$$2x^2 - 10x + 12 = 0$$

$$x^2 - 5x + 6 = 0$$

$$(x-2)(x-3) = 0$$

$$x = 2$$

$$x = 3$$

$$y = 1$$

$$y = 2$$

$\therefore z = 2 + i$  or  $3 + 2i$  (1)

Q(10b)

$$i) S_n = 1 + e^{i\theta} + e^{i2\theta} + e^{i3\theta} + \dots + e^{in\theta}$$

This is a GP with

$$a = 1 \quad r = e^{i\theta} \quad n = n+1$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1(e^{i\theta(n+1)} - 1)}{e^{i\theta} - 1}$$

$$= \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1}$$

Almost everyone realised it was the sum of a GP but I needed some WORKING  
 $\rightarrow$  either the sum of a GP formula or  $a = r =$  and  $n =$   
 Otherwise I could NOT award the mark.

ii) Take out  $e^{i(\frac{n+1}{2})\theta}$  and  $e^{-i\frac{\theta}{2}}$  as common factors

$$S_n = e^{i(\frac{n+1}{2})\theta} (e^{i(\frac{n+1}{2})\theta} - e^{-i(\frac{n+1}{2})\theta})$$

Very poorly done. Some students went to De Moivre  
 Only I succeeded (this way)

$$e^{i\frac{\theta}{2}} (e^{i\frac{\theta}{2}} - e^{-i\frac{\theta}{2}}) \quad \textcircled{1}$$

$$\text{Now } z + \frac{1}{z} = 2i \sin \theta$$

$$= \frac{e^{i(\frac{n+1}{2})\theta} (2i \sin(\frac{n+1}{2}\theta))}{e^{i\frac{\theta}{2}} (2i \sin \frac{\theta}{2})} \quad \textcircled{1}$$

$$\therefore S_n = \frac{e^{i(n+1)\frac{\theta}{2}} \sin\left(\frac{n+1}{2}\theta\right)}{e^{i\frac{\theta}{2}} \sin\frac{\theta}{2}}$$

$$(ii) \operatorname{Im}(1 + z + z^2 + \dots + z^n)$$

$$= \sin\theta + \sin 2\theta + \dots + \sin n\theta \quad (1)$$

from (ii)

About 15 or 20 students realised to equate imaginary parts & they all succeeded with this question.

$$S_n = \left( \cos \frac{n\theta}{2} + i \sin \frac{n\theta}{2} \right) \times \frac{\sin\left(\frac{n+1}{2}\theta\right)}{\sin\frac{\theta}{2}}$$

expand

$$(1) = \frac{\cos \frac{n\theta}{2} \sin\left(\frac{n+1}{2}\theta\right)}{\sin\frac{\theta}{2}} + i \frac{\sin \frac{n\theta}{2} \sin\left(\frac{n+1}{2}\theta\right)}{\sin\frac{\theta}{2}}$$

$$\operatorname{Im}(S_n) = \frac{\sin \frac{n\theta}{2} \sin\left(\frac{n+1}{2}\theta\right)}{\sin\frac{\theta}{2}}$$

$$\therefore \sin\theta + \sin 2\theta + \dots + \sin n\theta = \frac{\sin \frac{n\theta}{2} \sin\left(\frac{n+1}{2}\theta\right)}{\sin\frac{\theta}{2}}$$

Q11) let  $z^5 = r^5 \cdot \text{cis } 5\theta$

$$1 = 1 \text{ cis } \frac{\pi}{2}$$

$$r^5 = 1$$

$$\therefore r = 1$$

$$5\theta = \frac{\pi}{2} + 2k\pi$$

$$\theta = \frac{\pi}{10} + \frac{2k\pi}{5}$$

$$= \frac{\pi + 4k\pi}{10}$$

(1)

where  $k = 0, \pm 1, \pm 2$

$$z_1 = \text{cis } \frac{\pi}{10}$$

$$z_2 = \text{cis } \frac{\pi}{2} = i$$

$$z_3 = \text{cis } \frac{9\pi}{10}$$

$$z_4 = \text{cis } \left(-\frac{3\pi}{10}\right)$$

$$z_5 = \text{cis } \left(-\frac{7\pi}{10}\right)$$

(1)

Well done.

ii)  $z^5 - 1 = 0$  can be expressed as

$$(z - i) \left( z - \cos \frac{\pi}{10} \right) \left( z - \cos \frac{9\pi}{10} \right) \quad (1)$$

$$\left( z - \cos \frac{-3\pi}{10} \right) \left( z - \cos \frac{-7\pi}{10} \right)$$

Now,  $\alpha = \cos \frac{\pi}{10}$  and  $-\bar{\alpha} = \cos \frac{9\pi}{10}$  (1)

notice:

many students did not realise that

$$\beta = \cos \frac{-3\pi}{10} \text{ and } -\bar{\beta} = \cos \frac{-7\pi}{10}$$

$$\therefore (z - i) (z - \alpha) (z + \bar{\alpha}) (z - \beta) (z + \bar{\beta}) = 0$$

$$(z - i) [z^2 + (\bar{\alpha} - \alpha)z - \alpha\bar{\alpha}]$$

$$[z^2 + (\bar{\beta} - \beta)z - \beta\bar{\beta}] = 0$$

$$\text{Now } \alpha\bar{\alpha} = 1 \quad \beta\bar{\beta} = 1 \quad (1)$$

and

$$\bar{\alpha} - \alpha = \cos \frac{\pi}{10} - i \sin \frac{\pi}{10} - \cos \frac{\pi}{10} - i \sin \frac{\pi}{10}$$

$$= -2i \sin \frac{\pi}{10}$$

The more successful students wrote out the expansion. (1)

Similarly

$$\bar{\beta} - \beta = -2i \sin \frac{-3\pi}{10}$$

$$= 2i \sin \frac{3\pi}{10}$$

$$\therefore (z-i) \left( z^2 - 2 \sin \frac{\pi}{10} z - 1 \right)$$

$$(z^2 + 2 \sin \frac{3\pi}{10} z - 1)$$

Poorly done

as required  
All steps must be shown to obtain full marks.

$$(ii) (z-i) (z^4 + iz^3 - z^2 - iz + 1)$$

$$= (z-i) \left( z^2 - 2i \sin \frac{\pi}{10} z - 1 \right)$$

$$(z^2 + 2i \sin \frac{3\pi}{10} z - 1)$$

equate coefficients of  $z^2$

$$-1 = -2i \sin \frac{\pi}{10} \cdot 2i \sin \frac{3\pi}{10} - 1 - 1$$

$$1 = -4 i^2 \sin \frac{\pi}{10} \sin \frac{3\pi}{10}$$

$$\sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$$

Generally well done.

Please show working explicitly!